Twin Layer: Physical Simulation & Adversary Specifications

This reference documents the physical formulas, diurnal kinetic curves, incident engine dynamics, and adversary state machines powering the simulation engine in `twin/simulation.py`.

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1. Simulation Engine Principles & Dual-Clock System

1. **Dual-Clock Coordination:**
2. **Simulation Time:** Strictly deterministic RFC 3339 UTC timestamps (`YYYY-MM-DDTHH:MM:SSZ`).
3. **Wall-Clock Pacing:** Cycle pacing runs strictly at 10 Hz (100 ms per wall cycle) via `SimulationClock.pace_wall_clock()`. Wall-clock time is never leaked into Kafka payloads, SQLite truth readings, or MCP responses.
4. **Simulation Speed Acceleration (`SIM_SPEED`):**
5. Configurable via environment variable `SIM_SPEED` (default `1.0` , allowable range `0.1` to `60.0`).
6. Each simulation tick advances simulation time by $\Delta t = 0.1 \times \text{SIM_SPEED}$ simulation seconds.
7. All physical differential equations, noise terms, and exponential lags are written per simulation second, ensuring physical trajectory invariance regardless of `SIM_SPEED` .
8. **Euler-Maruyama Stochastic Integration & Lags:**
9. Noise terms scale with $\sqrt{\Delta t}$ in accordance with the Euler-Maruyama scheme: $\sigma \sqrt{\Delta t} \cdot \mathcal{N}(0, 1)$.
10. Dynamic inertia follows first-order exponential lag: $\alpha = 1 - \exp(-\Delta t / \tau)$, where τ is the characteristic time constant in simulation seconds.
11. **Dedicated Physics PRNG (`physics_rng`):**
12. Physics updates, camera variety factors, and stochastic Poisson arrival processes derive exclusively from a dedicated instance of `random.Random(SIM_SEED)` .
13. Network interactions, mock clients, and API requests do not advance the physics RNG, guaranteeing bit-for-bit trajectory determinism across identical runs.
14. **Municipal Zones & Geographic Coordinates:**
15. The city comprises 42 IoT devices distributed evenly (7 devices per zone) across 6 municipal zones.
16. Zone-2 ("Port & Coastal Harbor") is positioned at the residential center of **La Goulette** with centroid coordinates `(36.8185, 10.3050)` .
17. The 7 devices in each zone follow the deterministic geographic offset formula:
    $$\text{lat} = \text{round}(z_{\text{lat}} + (\text{idx} \times 0.0012) - 0.0036, 6)$$
    $$\text{lon} = \text{round}(z_{\text{lon}} + (\text{idx} \times 0.0015) - 0.0045, 6)$$

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2. Sensor Kinetics & Physical Formulations

The digital twin models 42 IoT devices across 5 municipal categories:

A. Traffic Camera (`traffic_camera`)

- **Metric:** Vehicle passage throughput in vehicles per minute (V).
- **Operating Bounds:** $0.0 \le V \le 200.0\text{ veh/min}$.
- *Diurnal Target Demand ($V^(t)$):*
  For simulation hour $t \in [0.0, 24.0)$:
  $$V^(t) = 25.0 + 95.0 \cdot \exp\left(-\frac{(t - 8.5)^2}{2 \cdot 1.2^2} \right) + 110.0 \cdot \exp\left(-\frac{(t - 17.5)^2}{2 \cdot 1.5^2} \right)$$

- Morning rush hour peaks around 08:30 (~120 veh/min).
- Evening rush hour peaks around 17:30 (~135 veh/min).
- Overnight baseline remains below 35 veh/min (~25 veh/min).
- **Per-Camera Variety:** Each traffic camera is assigned a fixed variety multiplier drawn from $[0.85, 1.15]$ using `physics_rng`.
- **Actuation Impact (`set_signal_plan`):**
- `normal` : Multiplier 1.0.
- `green_wave` : Multiplier 1.15 (clears congestion corridors).
- `red_restriction` : Multiplier 0.55 (dampens incoming vehicle density).
- `all_red` : Multiplier 0.05 (emergency cordon stop).
- `adaptive` : Targets the zone-wide mean target across both cameras in the zone.
- **Incident Impact (`traffic_accident`):**
- Reduces target throughput to $\times 0.3$ ramped over 60 simulation seconds.
- **Dynamics:** Time constant $\tau = 60\text{ s}$, noise $\sigma = 0.6\sqrt{\Delta t}\text{ veh/min}$. Steady-state spread $\sigma\sqrt{\tau/2} \approx 3.29\text{ veh/min}$.

B. Water Level Float (`water_level`)

- **Metric:** Drainage channel flood level in meters (H).
- **Operating Bounds:** $0.0 \le H \le 12.0\text{ meters}$.
- **Differential Equation:**
$$\frac{dH}{dt} = Q_{\text{in}}(t) - k \cdot c \cdot \sqrt{2g \max(0, H - H_0)}$$
- **Constants & Parameters:**
- Gravitational acceleration: $g = 9.81\text{ m/s}^2$.
- Retention threshold: $H_0 = 1.2\text{ meters}$.
- Calibrated discharge coefficient: $c = 0.00089$.
- Inflow rate: $Q_{\text{base}} = 0.003\text{ m/s}$ modulated by a $\pm 10\%$ tidal term with a 12.4-hour period ($44,640\text{ s}$):
$$Q_{\text{base}}(t) = 0.003 \cdot \left(1.0 + 0.10 \cdot \sin\left(\frac{2\pi \cdot t_{\text{sim_sec}}}{44640}\right)\right)$$
- Each device has a fixed catchment multiplier $C_i \sim \mathcal{U}(0.85, 1.15)$ drawn from the physics RNG applied to inflow ($Q_{\text{in}, i}(t) = Q_{\text{base}}(t) \cdot C_i$).
- Each device has a starting level drawn from the physics RNG ($H_{0, i} \sim \mathcal{U}(3.0, 3.8)\text{ meters}$).
- No sensor noise on water (spec preserved), determinism for a given seed, storms scale per zone ($Q_{\text{in}, i}(t) = Q_{\text{base}}(t) \cdot C_i + Q_{\text{base}}(t) \cdot (\text{intensity} - 1.0)$), and the two sensors in one zone stay strongly correlated.
- **Actuation Impact (`set_valve`):**
- Valve ratio k: `open` (1.0), `half` (0.5), `closed` (0.0), `"NN%"` ($\text{NN}/100$).
- Default initial valve position: `"half"` ($k = 0.5$).
- **Calibrated Behaviors:**
- *Nominal Half Valve:* Holds steady between 2.5 m and 4.5 m for 2+ simulation hours.
- *Closed Valve:* Starting from 3.5 m, crosses the 8.0 m critical threshold after 15 to 40 simulation minutes (~25 min).
- *Open Valve:* Rapid gravity drainage settles near $1.78\text{ m} \approx 1.8\text{ m}$.
- *Rain Storm ($\times 3$ Inflow) with Half Valve:* Crosses 8.0 m within 8 to 25 simulation minutes (~16.6 min).
- *Rain Storm ($\times 3$ Inflow) with Open Valve:* Equilibrium settles at $H \approx 6.41\text{ m}$, staying safely below 8.0 m.

C. Power Load Meter (`power_load`)

- **Metric:** Substation electrical load in megawatts (P).
- **Operating Bounds:** $5.0 \le P \le 120.0\text{ MW}$.
- **Diurnal Target Demand ($P^(t)$):*
- **Evening Transition ($20:00 \le t < 21:00$):**
  $$P^*(t) = 30.0 + 14.0 \cdot (1.0 + \cos(\pi(t - 20.0)))$$
  Smooth cosine descent from 58.0 MW down to 30.0 MW.
- **Deep Night ($21:00 \le t < 05:00$):**
  With $t_{\text{night}} = (t - 21.0) \pmod{24}$:
  $$P^*(t) = 30.0 + 3.0 \cdot \sin\left(\frac{2\pi \cdot t_{\text{night}}}{8.0}\right)$$
  Night load oscillates gently between 27.0 MW and 33.0 MW (strictly within $25 - 35\text{ MW}$).
- **Morning Transition ($05:00 \le t < 06:00$):**
  $$P^*(t) = 30.0 + 14.0 \cdot (1.0 - \cos(\pi(t - 5.0)))$$
  Smooth cosine ascent from 30.0 MW up to 58.0 MW.
- **Daytime ($06:00 \le t \le 20:00$):**
  $$P^*(t) = 30.0 + 70.0 \cdot \left(0.6 \cdot \sin\left(\frac{\pi(t - 6.0)}{14.0}\right) + 0.4\right)$$
  Daytime peak reaches 100.0 MW at 13:00 (within $85 - 110\text{ MW}$).
- Seamless continuity at boundary hours $05:00$, $06:00$, $20:00$, and $21:00$ prevents jumps $> 3\text{ MW}$ between consecutive simulation minutes (max step $< 0.75\text{ MW/min}$).
- **Actuation Impact (`switch_grid`):**
- Grid state factors: `grid_a` (1.0), `grid_b` (0.85), `island` (0.60), `shed` (0.40), `standby` (0.20).
- **Incident Impact (`grid_overload`):**
- Target demand is set to $\max(P^*(t) \cdot 1.35, 118.0\text{ MW})$, ensuring the overload is visible and exceeds the 114.0 MW threshold at any hour of day or night.
- If true power exceeds 114.0 MW continuously for 60 simulation seconds, **blackout** triggers: power target drops to 10% of normal target and emergency calls surge ($+4.0$).
- **Dynamics:** Time constant $\tau = 120\text{ s}$, noise $\sigma = 0.3\sqrt{\Delta t}\text{ MW}$. Steady-state spread $\sigma\sqrt{\tau/2} \approx 2.32\text{ MW}$.

D. Air Quality Station (`air_quality`)

- **Metric:** Ambient Air Quality Index (AQI).
- **Operating Bounds:** $10.0 \le AQI \le 350.0$.
- **Target Formulation:**
  $$\text{Target} = \text{Background}(\text{zone}) + 0.45 \cdot \text{mean}(\text{zone true traffic}) + \text{IndustrialTerm} + \text{IncidentTerms}$$
- **Parameters:**
- Zone Backgrounds: `zone-1` (18.0), `zone-2` (15.0), `zone-3` (20.0), `zone-4` (8.0), `zone-5` (14.0), `zone-6` (12.0).
- Clean Baseline: In `zone-4` (Green Hills), background of 8.0 yields measured AQI of about 19 at night and about 60 at rush hour.
- Industrial Contribution: `zone-3` receives $+25.0\text{ AQI}$.
- Incidents:
  - `fire` : $+150.0\text{ AQI}$ ramping linearly over 120 simulation seconds.
  - `air_hazard` : $+100.0\text{ AQI}$, mitigated by signal plans (`red_restriction` reduces bonus by 50%, `all_red` reduces bonus by 80%).
- **Dynamics:** Time constant $\tau = 300\text{ s}$, noise $\sigma = 0.15\sqrt{\Delta t}\text{ AQI}$. Steady-state spread $\sigma\sqrt{\tau/2} \approx 1.84\text{ AQI}$.

E. Emergency Call Dispatch (`emergency_call`)

- **Metric:** Municipal emergency 911/112 call volume in calls per minute (C).
- **Operating Bounds:** $0.0 \le C \le 30.0\text{ calls/min}$.
- **Arrival Process:**
- In each tick, arrivals $n \sim \text{Poisson}(\lambda \cdot \Delta t / 60)$ are drawn from `physics_rng`.
- Rate parameter λ (calls/min) is dynamic per zone:
  $$\lambda = 2.0 + 4.5 \cdot \mathbb{I}(\text{any zone cam} > 120) + 8.0 \cdot \mathbb{I}(\text{any zone water} > 8.0) + \text{IncidentSurges} - 1.0 \cdot N_{\text{generic_dispatches}}$$
  Floored at $\lambda_{\text{min}} = 0.2\text{ calls/min}$.
- Incident surges: `traffic_accident` ($+6.0$), `fire` ($+8.0$), `grid_overload` with blackout ($+4.0$).
- **Telemetry Reporting:**
- Reported reading is the exact count of arrivals during the past 60 simulation seconds, maintained in a FIFO deque per zone and capped at 30 calls/min.
- Active generic dispatches relieve emergency pressure for `DISPATCH_DURATION_S` (default 900 s).

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3. Physical Incident Engine

Physical incidents represent real municipal emergencies, distinct from adversary cyber attacks. They alter physical reality directly and can be resolved through timely city actuation.

Incident Type	Scope	Physical Effect	Resolution & Duration Defaults
`rain_storm`	Zone	Upstream inflow Q_{in} multiplied by intensity ($1.0 - 5.0$, default 3.0).	Defaults to `duration_s = 1800` (30 min) when omitted; naturally expires.
`traffic_accident`	Device (Camera)	Camera target multiplied by $\times 0.3$ ramped over 60 s; zone $\lambda + 6.0$.	Lasts until resolved by `dispatch` matching `traffic_accident`, `medical_emergency`, or `medical_surge` after 300 s response time (or cleared).
`fire`	Zone	AQI target $+150$ ramped over 120 s; zone $\lambda + 8.0$.	Lasts until resolved by `dispatch` matching `fire` after 180 s response time (or cleared).
`grid_overload`	Zone (Power)	Power target $\max(P^* \cdot 1.35, 118\text{ MW})$. Continuous load $> 114\text{ MW}$ for 60 s causes blackout (target $\times 0.10$, $\lambda + 4.0$).	Pre-blackout: resolved immediately by `switch_grid` to `grid_b`, `shed`, or `standby`. Post-blackout: resolved by `dispatch("blackout")` after 240 s. Lasts until resolved or cleared.
`air_hazard`	Zone	AQI target $+100$, mitigated by `red_restriction` or `all_red` signal plans.	Defaults to `duration_s = 1800` (30 min) when omitted; naturally expires.

Zero-Trust Isolation of Physical Incidents

- **MCP Invisibility:** Physical events are never exposed to City Brain or Guardian through MCP queries.

- **Contained / Attacked Commands:** Any command refused due to device containment (`DEVICE_CONTAINED`) or suppressed by an actuator attack does **not** resolve physical incidents.
- **Red Team API (Port 8100):** Protected by `X-Red-Team-Token` :
- `POST /api/event/inject` : Injects incident (`type` , `zone_id` , `device_id` , `intensity` , `duration_s`).
- `POST /api/event/clear` : Clears specific incident or all incidents.
- `GET /api/event/active` : Lists active physical events.
- **Ground Truth Ledger:** Initiations, durations, and resolutions are recorded in SQLite table `physical_events` .

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4. Adversary Capability Engine

The adversary model corrupts wire telemetry while the internal SQLite ledger preserves ground truth:

Attack Capability	Wire Manipulation Method	Detection & Guardian Mitigation
`device_physical`	Overwrites telemetry value with attacker-defined spoofing (`spoofed_value` or `value` parameter); generates valid Ed25519 signature with legitimate device key.	Guardian flags anomalous physical gradients or cross-sensor contradictions; invokes `rollback_reading` or `isolate_device` .
`device_key`	Attacker extracts valid private key and broadcasts malicious readings (`spoofed_value` or `value` parameter) that cryptographically verify on wire.	Guardian cross-checks with historical sensor baselines and neighbor correlations; invokes `quarantine_device` .
`link`	Flips payload bits (`corrupt_bytes`), drops readings (`drop`), replays past packets (`replay`), or buffers frames in a FIFO delay queue (`delay`). When cleared, held frames are immediately released in FIFO order.	Guardian detects invalid Ed25519 signatures, sequence gaps, or timestamp regression; alerts operators.
`rogue`	Broadcasts telemetry under an unregistered identifier (e.g., `rg-01`).	Guardian compares against `list_devices` ; quaranines rogue hardware via MCP `quarantine_device` .
`actuator`	Acknowledges MCP commands as `"ok"` but leaves true physical state unchanged.	City Brain observes telemetry failing to converge toward expected physical setpoints.
`sensor_fault`	Injects subtle linear drift ($\Delta t \cdot \text{drift_rate}$), stuck frozen values, or zero dropouts.	Statistical moving-average and deviation filters identify sensor drift.

4.1 Link Adversary and Delay Backlog Lifecycle

When the `link` capability is configured in `delay` mode (`delay_ticks`), frames for the targeted device are held in a FIFO delay queue up to a maximum depth of 600 ticks. The count of unreleased frames across all device buffers is exposed in the Red Team API (`/api/status`) under `delayed_frames_pending` . When a link attack in delay mode is cleared (via `/api/attack/clear` , individual attack clear, or simulator settle), all held frames are immediately released to the wire in strict chronological order instead of dribbling out over subsequent ticks, and `delayed_frames_pending` drops to 0.

4.2 Parameter Aliases for Injection Payloads

For the `device_physical` and `device_key` adversary capabilities, the target sensor value may be specified using either `spoofed_value` or `value` in the injection parameter dictionary (`params: {"spoofed_value": 185.0}` or `params: {"value": 185.0}`). If both or either are provided, the simulator resolves and normalizes them identically to enforce consistent spoofing on wire telemetry.